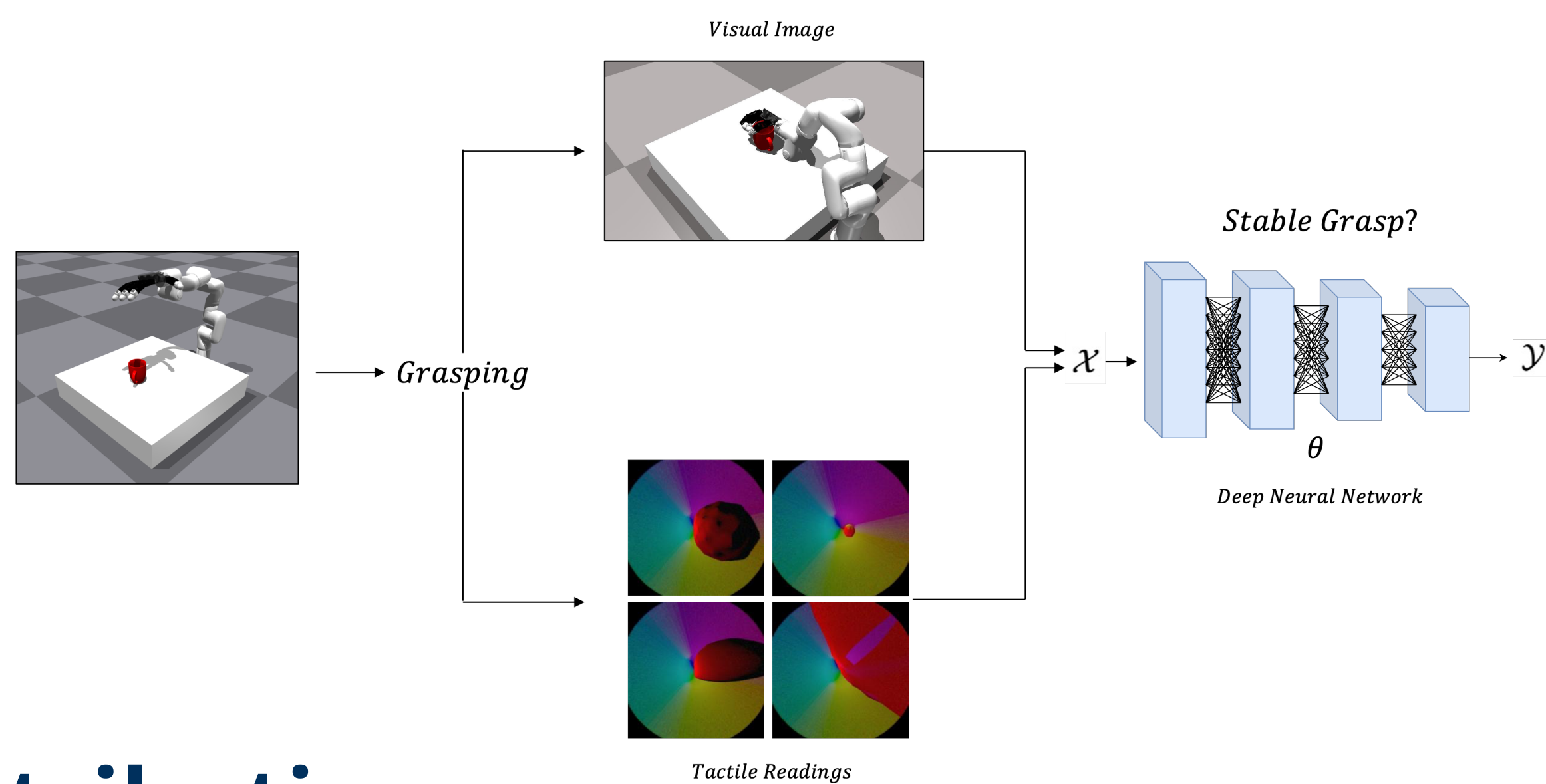




Motivation

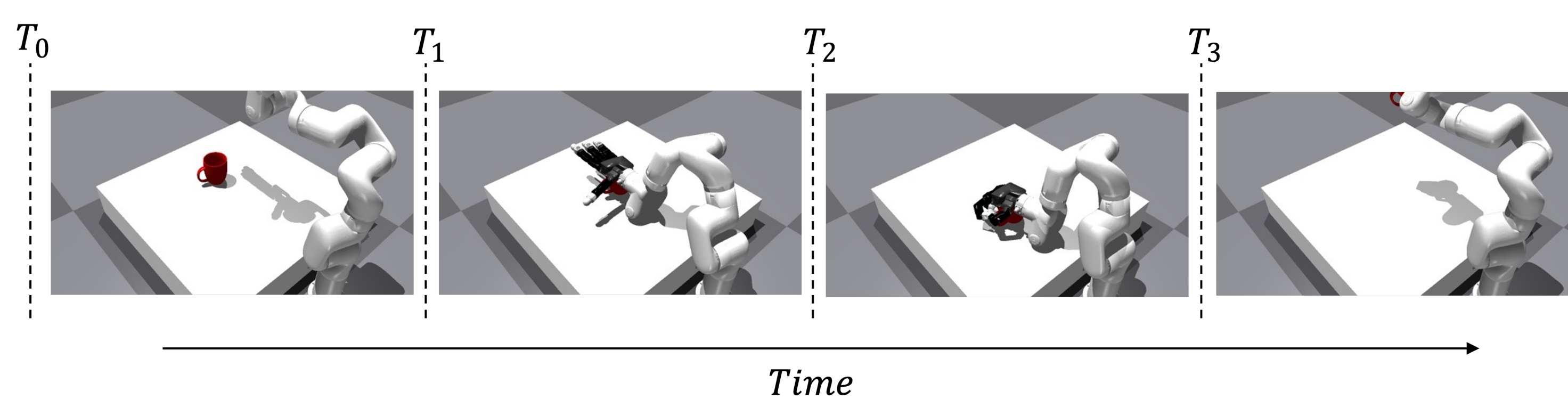
- Mastering the art of grasping is crucial for robots to become a versatile tool for our daily lives.
- For a single grasp with a multi-finger hand, a human makes use of different perception factors, including vision and tactile sensing.
- In the field of robotics, the sense of vision is primarily used for solving various stages of robot grasping, even though vision is only an indirect measurement.
- Does the additional use of tactile sensing as form of direct measurement aid in predicting the stability of a multi-finger grasp?



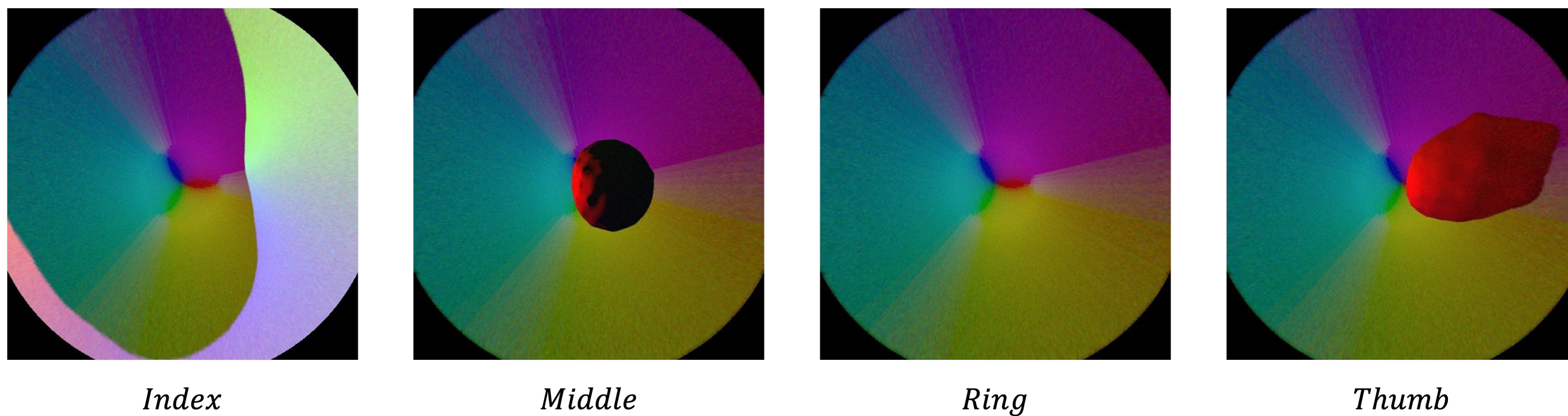
Contribution

- We propose an end-to-end visuo-tactile model to learn the grasp stability of a given multi-finger grasp in simulation.
- We compare the visuo-tactile model against different models that make use of a single modality. Our results show that the visuo-tactile models outperforms single modality models in terms of accuracy.

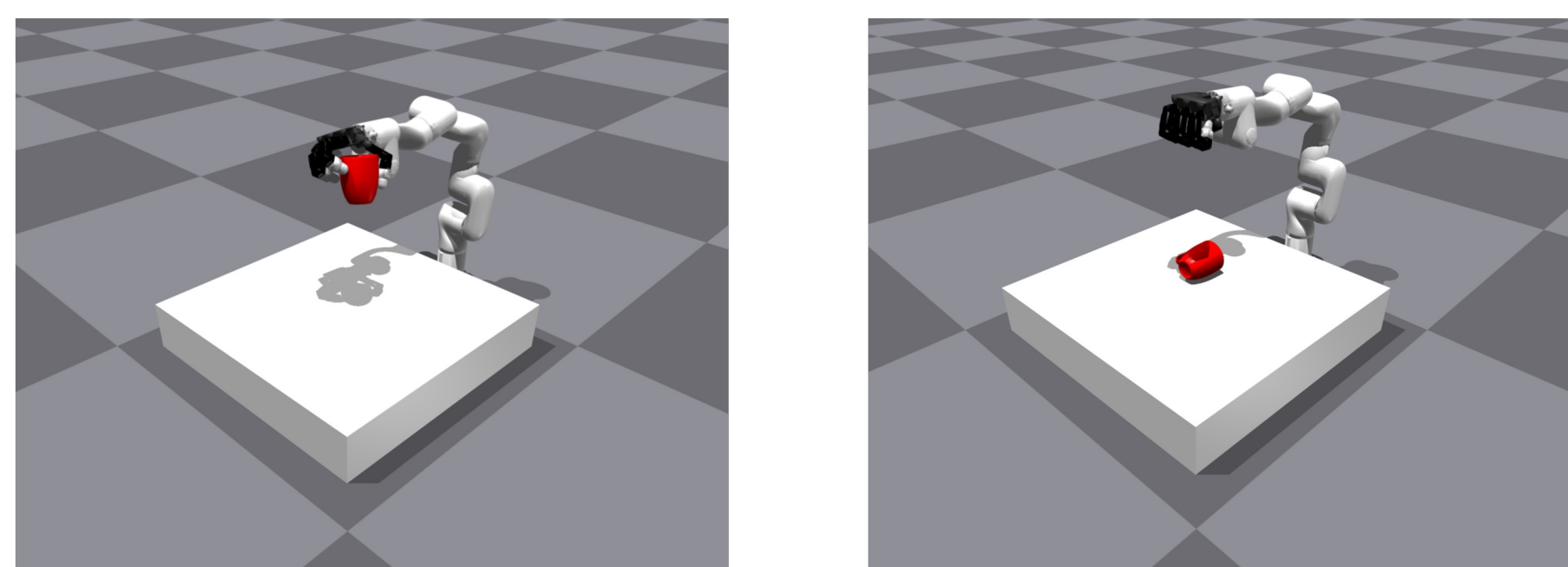
Data Collection



- Visual data** is collected at four distinct time steps using an over-the-shoulder RGB camera inside the simulation environment.



- Tactile data** is collected after the hand established contact to the object (**T₂**). The data originates from the simulation of an omnidirectional high-resolution vision-based tactile sensor.



- A grasp is considered **stable** if the hand still holds the object after returning to its initial position.
- A **binary label** $l \in \{0, 1\}$ is indicating the success (left, $l = 1$) or failure (right, $l = 0$).

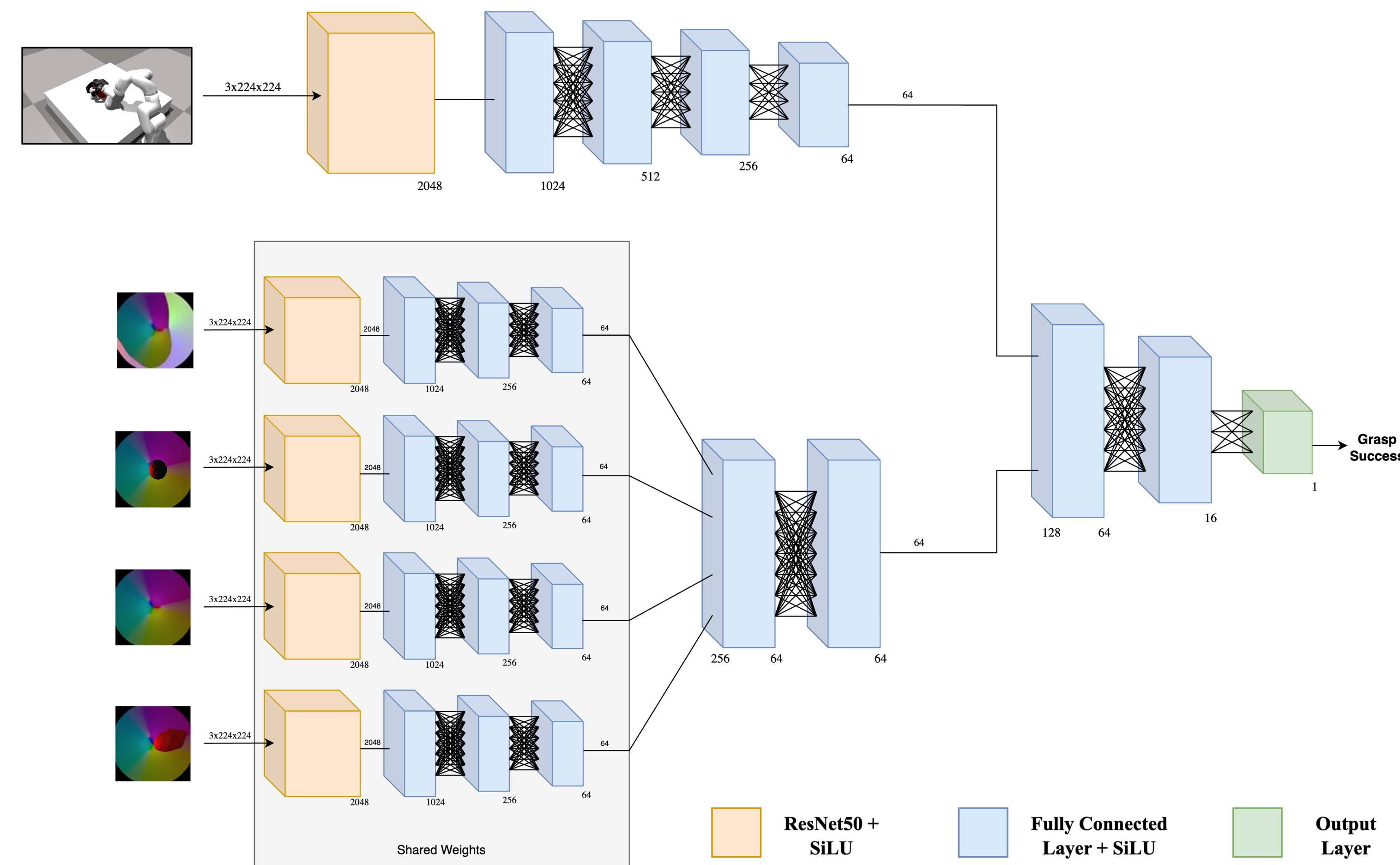
References

[1] Roberto Calandra, Andrew Owens, Manu Upadhyaya, Wenzhen Yuan, Justin Lin, Edward H Adelson, and Sergey Levine. "The feeling of success: Does touch sensing help predict grasp outcomes?" In: arXiv preprint arXiv:1710.05512 (2017)

[2] Shaoxiong Wang, Mike Lambeta, Po-Wei Chou, and Roberto Calandra. "Tacto: A fast, flexible, and open-source simulator for high-resolution vision-based tactile sensors". In: IEEE Robotics and Automation Letters 7.2 (2022), pp. 3930–3937

[3] Yasemin Bekiroglu. "Learning to assess grasp stability from vision, touch and proprioception". PhD thesis. KTH Royal Institute of Technology, 2012

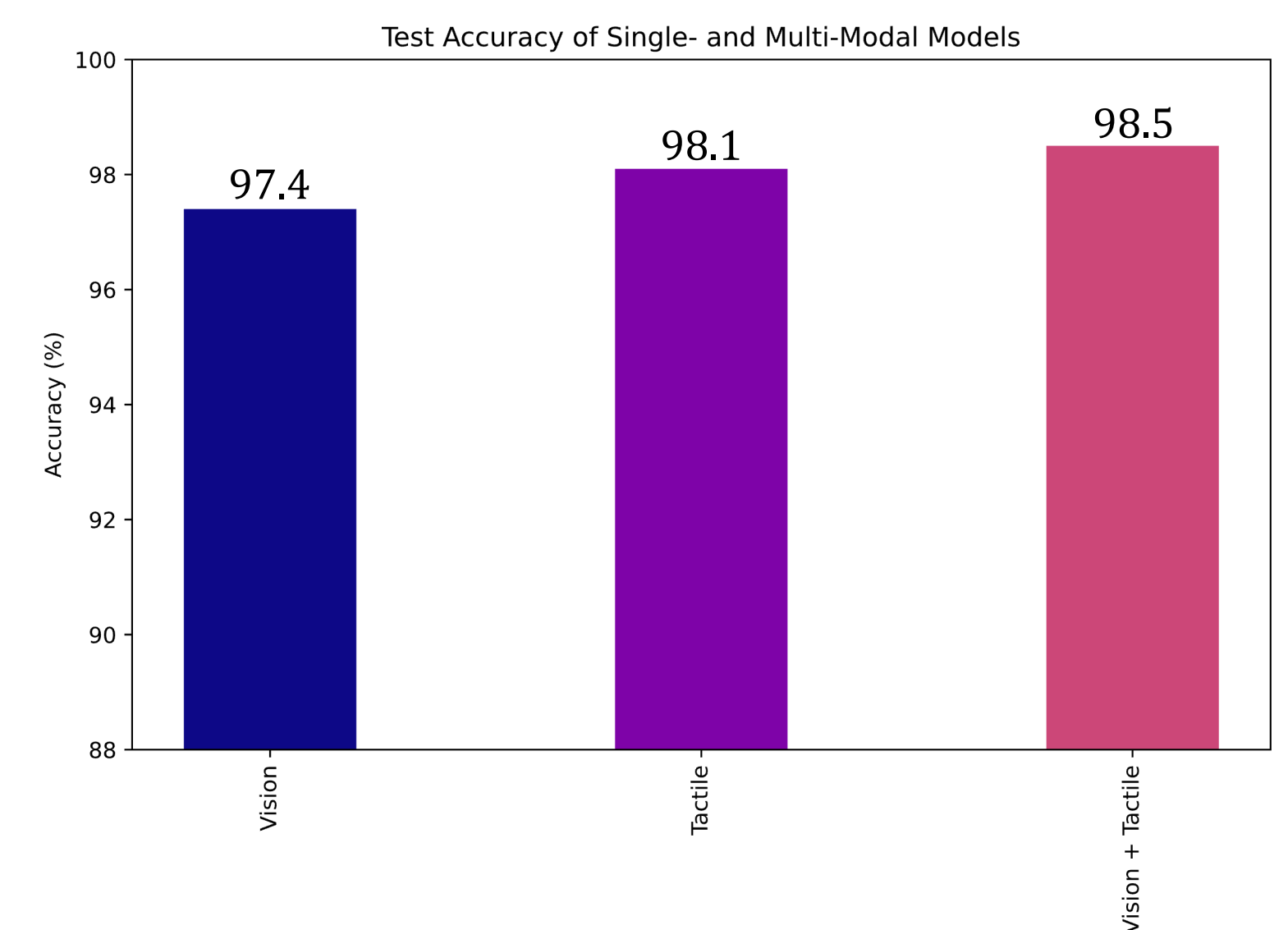
Stability Prediction Model



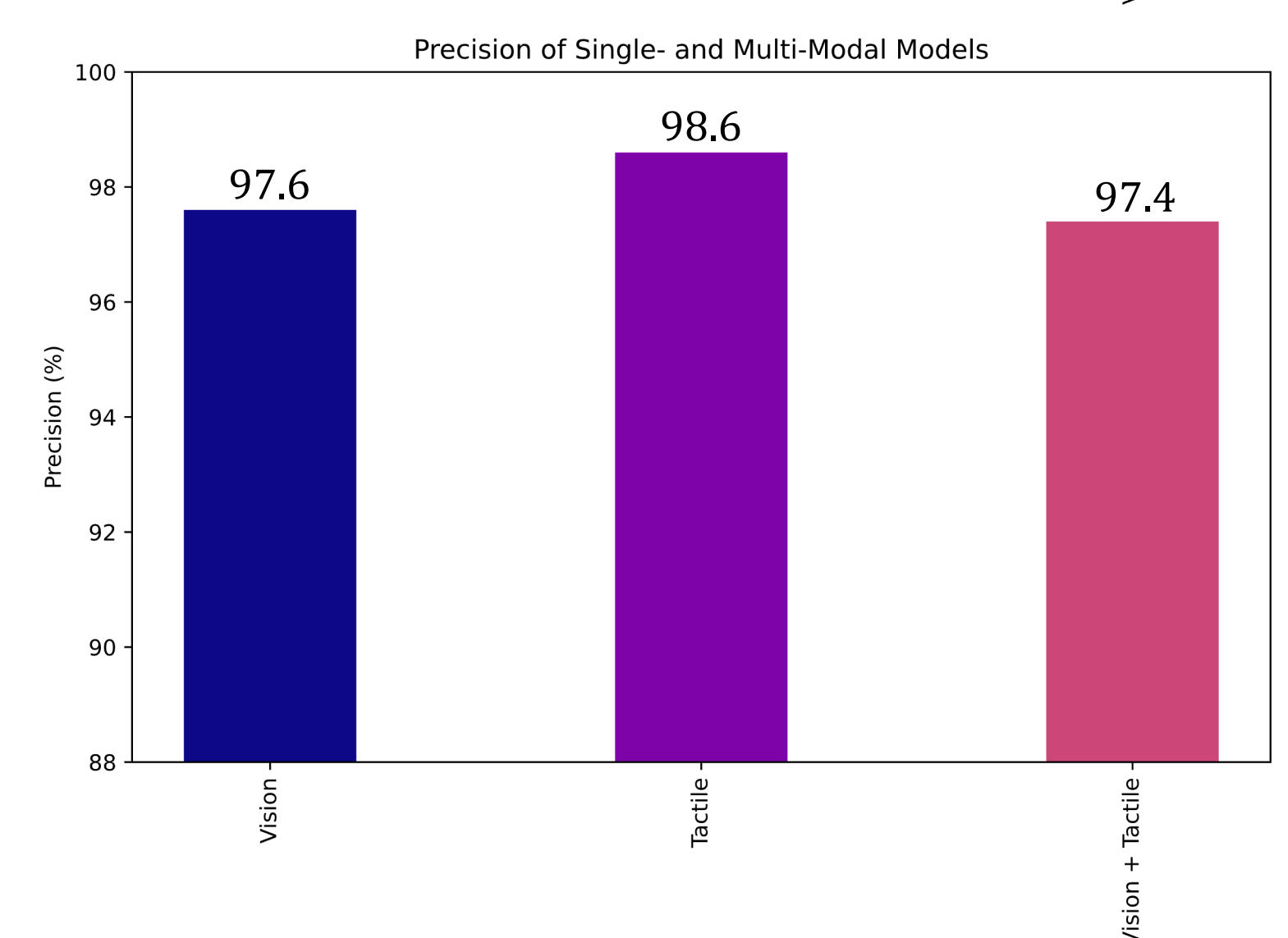
- Visuo-tactile model** leveraging visual and tactile data to predict the stability of a grasp as the probability of a successful object lifting.
- To learn the tactile features, we propose the concept of **shared weights**.
- The features of the single modalities are fused in an **intermediate stage**.

Experimental Results

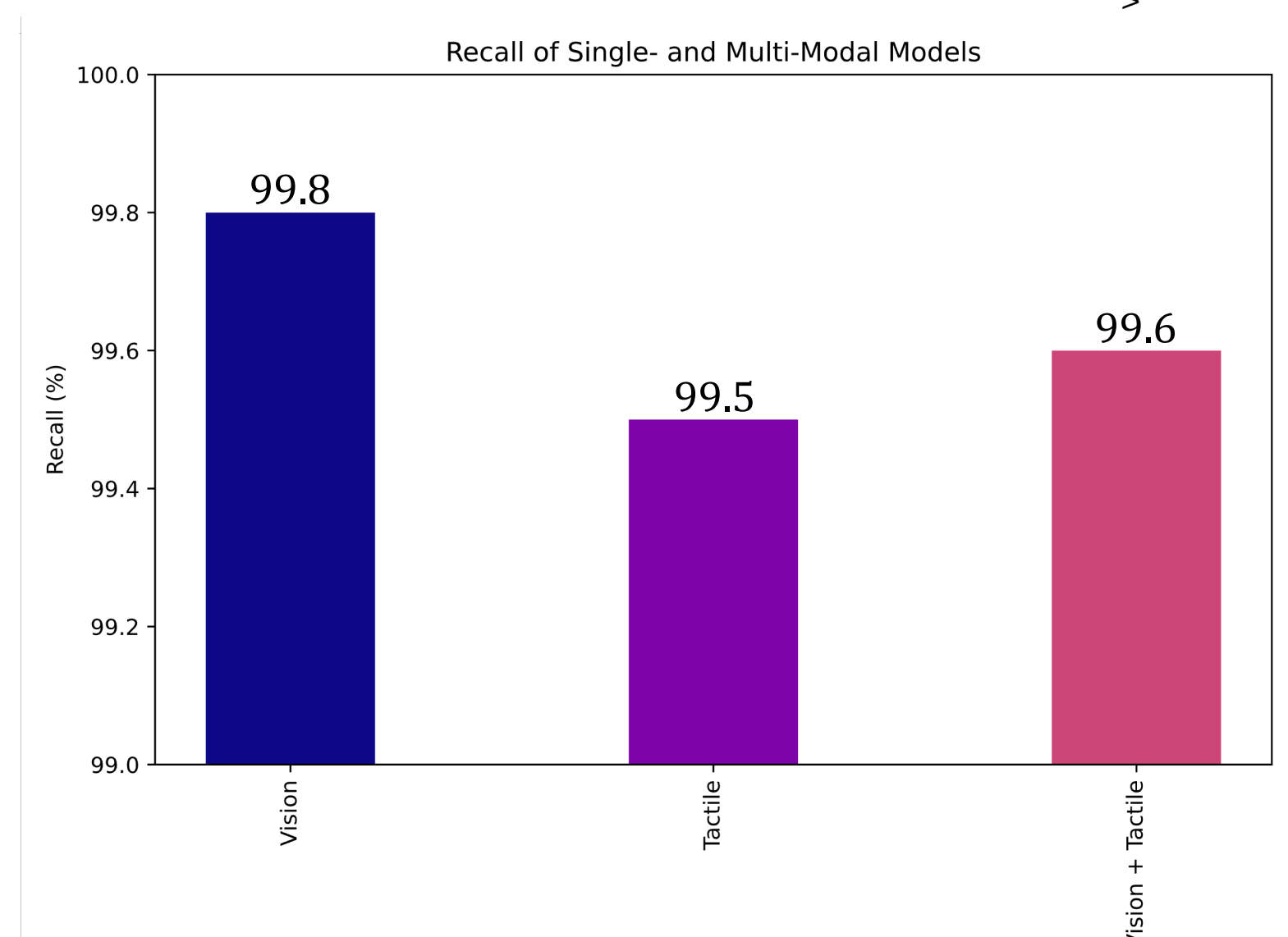
- Visuo-tactile model outperforms the vision and tactile only model in terms of accuracy by up to 1.1% on an unseen test dataset.



- The tactile only models shows the highest precision whereas the visuo-tactile models lacks in precision.



- The vision only model shows the highest recall whereas the tactile only model performs worst regarding this value



- Results indicate the **beneficial use of tactile sensing for stability prediction**.

Conclusion & Future Work

- We proposed models using visual and tactile data from a vision-based omnidirectional tactile sensor to predict grasp stability.
- We showed that tactile sensing helps in predicting the stability of a multi-finger grasp compared to purely vision-based methods.
- For the accuracy of different modality models, we showed that it holds: **Tactile + Vision > Tactile > Vision**.
- In future work we plan to increase the rate of successful grasps in the training set, use a more diverse set of graspable objects and evaluate the models on real world data.